

Practicing pandas: a complete example

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After presenting the logic of `Pandas` in the previous chapter, this chapter illustrates the features of the *package* using greenhouse gas emission data from `Ademe`.

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Table of contents

Informations additionnelles	1
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The `Pandas` part has recently evolved. You can find the content related to `Pandas` in the following chapters:

- Introduction to `Pandas` and the logic of the *DataFrame*;
- Combining data with `Pandas`: joins;
- Handling data with `Pandas`: statistics and reshaping;
- Pandas: limits and alternative ecosystems;

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via `uv`. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette",
    "contextily=1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal=3.11.4",
    "graphviz=0.20.3",
```

```

    "great-tables ≥ 0.12.0",
    "gt-extras ≥ 0.0.8",
    "ipykernel ≥ 6.29.5",
    "jupyter ≥ 1.1.1",
    "jupyter-cache ≥ 1.0.0",
    "kaleido ≥ 0.2.1",
    "langchain-community ≥ 0.3.27",
    "loguru = 0.7.3",
    "markdown ≥ 3.8",
    "nbcclient ≥ 0.10.0",
    "nbformat ≥ 5.10.4",
    "nlTK ≥ 3.9.1",
    "pandas ≥ 3.0",
    "pip ≥ 25.1.1",
    "plotly ≥ 6.1.2",
    "plotnine ≥ 0.15",
    "polars ≥ 1.8.2",
    "pyarrow ≥ 17.0.0",
    "pynsee ≥ 0.1.8",
    "python-dotenv ≥ 1.0.1",
    "python-frontmatter ≥ 1.1.0",
    "pywaffle ≥ 1.1.1",
    "requests ≥ 2.32.3",
    "scikit-image ≥ 0.24.0",
    "scikit-learn ≥ 1.8.0",
    "scipy ≥ 1.13.0",
    "seaborn ≥ 0.13.2",
    "selenium < 4.39.0",
    "spacy ≥ 3.8.4",
    "webdriver-manager ≥ 4.0.2",
    "wordcloud = 1.9.3",
]

[tool.uv.sources]
cartiflette = { git = "https://github.com/insee-fr/lab/cartiflette" }
gdal = [
    { index = "gdal-wheels", marker = "sys_platform == 'linux'", },
    { index = "geospatial-wheels", marker = "sys_platform == 'win32'", },
]

[[tool.uv.index]]
name = "geospatial-wheels"
url = "https://nathanjmcDougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean ≥ 4.0.1",
]

```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.